Replication: The Credit Default Swap–Bond Basis

Bai, Collin-Dufresne (2019)

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1 Introduction

This document presents a replication of Figure 1 and Table 1 from Bai and Collin-Dufresne's "The Credit Default Swap–Bond Basis" (2019). The paper examines the relationship between CDS spreads and bond-implied CDS spreads (PECDS) across different credit rating categories and crisis phases.

The CDS-bond basis is defined as the difference between the market CDS spread and the bond-implied CDS spread:

$$\text{Basis} = \text{CDS Spread} - \text{PECDS} \tag{1}$$

2 Background and Economic Context

2.1 The CDS-Bond Basis Arbitrage

2.1.1 The Theoretical Arbitrage

In a frictionless market, the CDS spread and bond-implied credit spread should be equal. When the basis is negative (CDS $<$ PECDS), an arbitrageur could theoretically:

1. Purchase the corporate bond
2. Finance the purchase via the repo market
3. Buy CDS protection on the bond
4. Lock in a risk-free profit equal to the basis

This arbitrage should eliminate any persistent negative basis.

2.1.2 Why the Arbitrage Failed: Limits to Arbitrage

However, during the 2007-2009 financial crisis, the basis became persistently and severely negative, suggesting the arbitrage was not risk-free. The paper investigates four key frictions:

- **Collateral Quality**: Bonds with lower credit ratings face higher haircuts in repo markets, requiring more risk capital
- **Liquidity Risk**: Illiquid bonds have higher transaction costs and risk of fire-sale losses
- **Counterparty Risk**: CDS protection is worthless if the protection seller defaults alongside the reference entity
- **Funding Liquidity Risk**: Arbitrageurs face mark-to-market losses when the basis widens as funding costs spike

3 Data Sources

The replication uses the following data sources:

- **CDS Data**: Markit CDS spreads from WRDS (specifically, we use par spreads to match the paper)
- **Bond Data**: WRDS Bond Returns database with monthly bond prices and yields
- **Bond Characteristics**: Mergent FISD (Fixed Income Securities Database) for bond issue details
- **Credit Ratings**: Mergent FISD ratings from S&P and Moody's
- **Treasury Data**: Monthly Treasury yields from WRDS used in PECDS approximation

4 Methodology

4.1 Data Preparation

Bonds are filtered according to the following criteria from the paper:

1. Remove convertible bonds
2. Keep only fixed or zero coupon bonds (no floating rate)
3. Time-to-maturity between 3 and 7.5 years (to match 5-year CDS liquidity)
4. Remove bonds with missing prices, yields, or company symbols
5. Keep only rated bonds (investment grade and high yield)
6. Keep only corporate bond types (debentures, MTNs, zero coupon)

CDS contracts are filtered to:

- USD currency only
- 5-year tenor (most liquid maturity)

The paper's sample period is July 2006 - December 2014. Our extended period is January 2015 - March 2022.

4.2 Crisis Phases

Following the paper, the sample is divided into four phases:

- **Before Crisis**: July 2006 - June 2007
- **Crisis I**: July 2007 - August 2008 (subprime crisis to Lehman bankruptcy)
- **Crisis II**: September 2008 - September 2009 (post-Lehman failure)
- **Post-crisis**: October 2009 - December 2014

4.3 Descriptive Summary of the Underlying Data

On the next page are some summary statistics for the underlying matched bond-CDS dataset and the key variables used in the analysis. Table 1 reports summary statistics for the main spread variables, while Figure 1 plots the average daily CDS spread and average daily bond-implied CDS spread proxy (PECDS) over time. Together, these descriptive statistics provide context for the replicated basis results by showing the typical correlation of the underlying spread series.

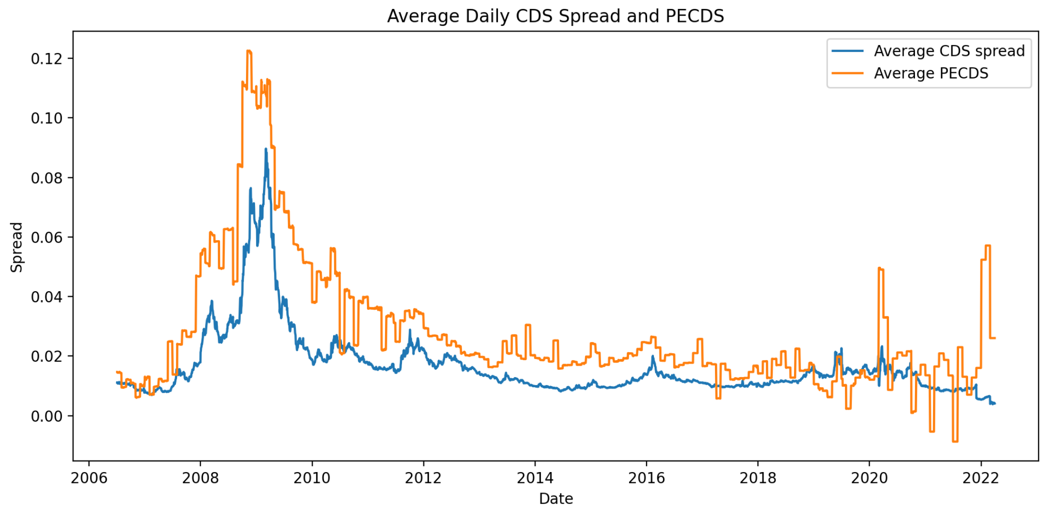

Figure 1: Average daily CDS spreads and PECDS over time in the sample. This figure helps motivate the basis analysis by showing the behavior of the two underlying spread series whose difference defines the CDS–bond basis. Periods where the two series diverge are the periods that generate large (absolute) basis values.

Table 1: Summary statistics of the underlying data. This table reports descriptive statistics for the key variables used in the analysis: the market CDS spread, the PECDS proxy constructed from bond and Treasury yields, the CDS–bond basis in basis points, and bond yields. The main takeaway here is to provide helpful intuition for interpreting the replicated results.

Variable	Mean	SD	P10	Median	P90	N
CDS spread	0.02	0.04	0.00	0.01	0.04	16727368
PECDS	0.03	0.05	-0.00	0.02	0.07	16727368
Basis (bps)	-135.02	427.46	-410.82	-110.58	158.32	16727368
Bond yield	0.04	0.05	0.00	0.03	0.08	16727368

4.4 Simplified PECDS Calculation

In the original paper, the bond-implied CDS spread (PECDS) is computed using bond prices together with the risk-free swap curve to bootstrap survival probabilities and construct a synthetic CDS spread implied by bond prices (they describe their methodology in Appendix A).

Implementing the methods used by Bai and Collin-Dufresne (2019) requires daily bond prices, a full swap rate term structure, and a detailed survival probability calibration. For our replication, those inputs are not available in the WRDS Bond Returns database, which provides only monthly bond prices and yields. Accordingly, we construct a simplified proxy for PECDS using bond yield spreads relative to Treasury securities. Specifically, for each bond observation we match the bond to a Treasury security with a similar maturity and compute

$$\text{PECDS}_{proxy} = y_{bond} - y_{Treasury} \tag{2}$$

where y_{bond} is the corporate bond yield and $y_{Treasury}$ is the yield of the closest-maturity Treasury security on the same date.

This approximation is extremely crude, but it was an attempt at improving upon simply using $\text{PECDS}_{proxy} = y_{bond}$, and is motivated by the general intuition that

$$y_{bond} \approx y_{risk-free} + \text{credit spread.} \tag{3}$$

Subtracting the Treasury yield therefore (in theory) isolates a rough estimate of the bond's credit spread. Since CDS spreads represent the cost of insuring against default, they should theoretically correspond to the credit spread "component" of the bond yield. Under this intuition, the yield spread relative to Treasuries provides a crude but somewhat reasonable proxy for PECDS.

5 Results

5.1 Figure 1 Replication

Figures 2 and 3 show the dispersion of the CDS–bond basis over time, separated by credit rating (i.e., investment-grade vs. high-yield bonds). The solid line represents the median basis, while the dashed lines show the 10th and 90th percentiles.

Figure 2: Dispersion of the CDS–bond basis by credit rating (replication window). Top panel shows investment-grade bonds; bottom panel shows high-yield bonds. Solid line: median. Dashed lines: 10th and 90th percentiles.

Our results are slightly off from the papers in terms of spread levels, but the general shape is roughly the same. Intuitively, our basis levels are not the same as the paper's because we do not calculate PECDS the same (i.e., as rigorously) as Bai and Collin-Dufresne (2019). That said, we see the same trend that during the global financial crisis, the basis becomes highly volatile, trending downward (i.e., becoming more negative) for both investment-grade and high-yield bonds.

Figure 3: Dispersion of the CDS–bond basis by credit rating (extended window). Top panel shows investment-grade bonds; bottom panel shows high-yield bonds. Solid line: median. Dashed lines: 10th and 90th percentiles.

In Figure 3 we see a similar pattern emerge: the basis becomes more negative in 2020 around the COVID-19 pandemic, a time of significant financial instability. This suggests that in addition to our replication, our extension captures the general idea that Bai and Collin-Dufresne (2019) are exploring in their paper (as financial instability worsens, the arbitrage relationship that should exist between CDS spreads and PECDS goes away).

5.2 Table 1 Replication

Table 2 presents summary statistics of the CDS-bond basis for the replication period (July 2006 - December 2014), matching the paper's sample. Table 3 shows results for the extended period (January 2015 - March 2022). Statistics include the mean, standard deviation (SD), 10th percentile (P10), and 90th percentile (P90).

Once again, we see the same trend as the figures: while the levels are not the exact same as the paper's, we see a similar pattern. Despite mean levels not varying significantly across periods in our replication, increased standard deviations in the crisis periods (particularly the Global Financial Crisis) are indicative of higher volatility in the basis. This is inline with the trends discussed above (as well as in the paper); broadly speaking, during crises, the arbitrage relationship between CDS spreads and PECDS becomes less stable.

Table 2: Summary statistics of discrepancies in CDS and cash bond spreads (Replication Period)

	Before Crisis July 2006–June 2007				Crisis I July 2007–Aug. 2008				Crisis II Sept. 2008–Sept. 2009				Post-crisis Oct. 2009–Dec. 2014			
	Mean	SD	P10	P90	Mean	SD	P10	P90	Mean	SD	P10	P90	Mean	SD	P10	P90
ALL	-27	192	-126	71	-206	259	-369	-19	-389	704	-791	32	-125	343	-328	143
IG	-26	175	-109	20	-198	216	-341	-83	-389	413	-650	-53	-143	280	-304	57
HY	-30	215	-175	157	-222	316	-465	63	-388	1114	-1241	697	-90	427	-408	280
AAA/AA	344	304	-52	727	243	317	-214	631	-217	753	-777	349	-246	521	-610	118
A	173	369	-78	650	-1	331	-253	483	-275	333	-543	16	-182	374	-453	47
BBB	-49	73	-106	14	-235	94	-340	-146	-370	403	-667	-87	-168	218	-440	66
BB	-66	145	-228	144	-248	193	-474	-46	-7	1576	-1234	1809	-86	111	-227	39
B	15	217	-167	233	-207	207	-413	21	40	1313	-1699	1709	-53	218	-325	196
CCC	-43	156	-196	93	86	402	-301	659	-173	1568	-2193	1745	174	537	-551	574

Notes: This table replicates Table 1 from Bai and Collin-Dufresne (2019), showing descriptive statistics for the average CDS-bond basis across four crisis phases. All entries are in basis points.

Table 3: Summary statistics of discrepancies in CDS and cash bond spreads (Extended Period)

	Extended Period Jan. 2015–Present			
	Mean	SD	P10	P90
ALL	-54	377	-207	164
IG	-69	196	-187	41
HY	6	635	-392	470
AAA/AA	-84	0	-84	-84
A	-43	105	-142	84
BBB	-75	73	-167	-3
BB	-99	30	-127	-68
B	-922	0	-922	-922
CCC	-4231	0	-4231	-4231

Notes: This table extends the analysis to the period January 2015 through present. All entries are in basis points.

5.3 Implementation Challenges

Several practical challenges arose during the replication process. One of the most significant difficulties involved the construction of the bond-implied CDS spread (PECDS). As discussed in the PECDS section, the methodology described in Bai and Collin-Dufresne (2019) requires daily bond prices and a complete swap rate term structure in order to bootstrap survival probabilities and generate a synthetic CDS spread from bond prices. Because the WRDS Bond Returns dataset provides only monthly bond yields, it was not possible to implement the full procedure described in Appendix A of the paper.

Initial attempts to approximate PECDS using the raw bond yield produced basis values that were substantially larger in magnitude than those reported in the paper (leading to significantly more negative basis levels). This outcome is expected because, as discussed above, bond yields embed both the risk-free rate and the credit spread component. To address this issue, we constructed a Treasury-adjusted proxy by subtracting a maturity-matched Treasury yield from the corporate bond yield. This adjustment is economically intuitive and isolates a rough measure of the bond's credit spread, and it led to substantial improvement in the empirical dispersion of the basis relative to the raw yield proxy. That said, our basis levels were still off from the paper's, highlighting the importance of the more sophisticated PECDS construction used in the original paper, which removes additional components such as liquidity premia, coupon effects, and term structure distortions.

Another implementation challenge involved the scale of the raw CDS dataset. The Markit CDS database contains tens of millions of observations, and an initial attempt to merge the entire CDS universe with the bond dataset resulted in a matched dataset exceeding 90 million rows, which caused memory constraints during processing. This issue was resolved by applying filters to the CDS dataset prior to matching, restricting the sample to USD-denominated five-year contracts, which are the most liquid instruments in the CDS market. This preprocessing step reduced the matched dataset to approximately 17 million observations and allowed the pipeline to run reliably.

One other thing worth noting is that the CDS dataset available through WRDS contains restructuring clauses labeled XR14 and XR, whereas the paper focuses on contracts with the Modified Restructuring (MR) clause. This discrepancy might reflect changes in CDS contract conventions over time and could be another explanation for the differences in spread levels relative to the original study.

6 Conclusion

In conclusion, despite the data and methodological limitations, the overall empirical patterns obtained in the replication are broadly consistent with the findings reported in Bai and Collin-Dufresne (2019). In both the replication and the original study, the CDS–bond basis remains close to zero prior to the financial crisis, becomes sharply negative during the crisis period, and gradually moves back toward zero in the subsequent recovery phase. Finally, our extension displays a simialr pattern with the COVID-19 pandemic, further illuminating the topic of the paper.